

STOCHASTIC METHODS IN WATER RESOURCES

Unit 3: Random functions

Lecture 9: Definitions, random function types, stationary random functions

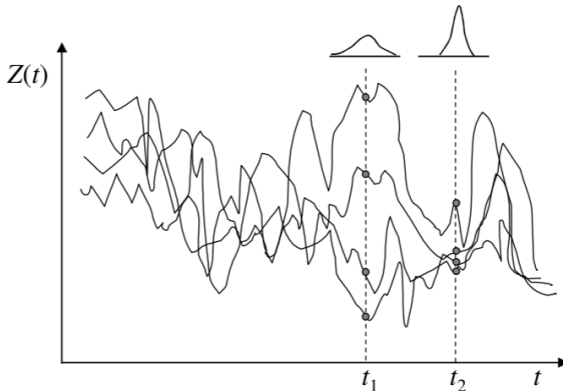
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Definitions

- ▶ Consider a **hydrological variable** Z that can either varies in space (e.g. hydraulic conductivity) or in time (e.g. surface water level) or in space and time (e.g. Groundwater depth).
- ▶ The value of any of these variables is known only in those places where it is observed, and it is unknown in other places.
- ▶ Accordingly, the value of Z is uncertain, to deal with this, instead of using one single function either in time ($Z(t)$), or space ($Z(x)$) or space and time ($Z(x, t)$), multiple functions are used (see figure for $Z(t)$).
- ▶ Each of these functions have the same probability of representing the true.
- ▶ The family of equally probable functions representing Z is known as an **ensemble** or a **random function**.



Definitions

- ▶ **Random field**: If Z is a function of space ($\mathbf{x}$), $Z(\mathbf{x})$, $\mathbf{x} = (x, y, z)$.
- ▶ **Random or stochastic process**: If Z is a function of time (t), $Z(t)$.
- ▶ **Space-time random field**: If Z is a function of space ($\mathbf{x}$) and time (t), $Z(\mathbf{x}, t)$, $\mathbf{x} = (x, y, z)$.

Here, we are going to focus on **random process** ($Z(t)$). Note that in the figure above, four functions $Z(t)$ (**ensemble members**) are shown and each of them are known as **realizations** and denoted as $z(t)$. The number of **realizations**, depending on the variable's nature, can be finite or infinite.

Another way of explaining this concept is to imagine a **random function** as a collection of **random variables** (one at every location in space or point in time). Observing at the figure above, for each time t , a $Z(t)$ is defined. This is why for each t , $Z(t)$ is described with a **pdf** $f_Z(z; t)$, which depends not only on the value of z but also on the value of t . These **pdfs** are estimated based on samples of data taken at particular times (e.g. t_1). As shown in the figure, the **pdfs** of the data sample at t_1 and t_2 are different, and indicate that variance is larger in t_1 than in t_2 and thus more uncertain. At each point in time (t), μ_Z is calculated as:

$$\mu_Z(t) = E[Z(t)] = \int_{-\infty}^{\infty} z f_Z(z; t) dz$$

and σ_Z^2 is:

$$\sigma_Z^2(t) = E[(Z(t) - \mu_Z(t))^2] = \int_{-\infty}^{\infty} (z - \mu_Z(t))^2 f_Z(z; t) dz$$

Definitions

The **random variables** are usually correlated in time (or in space for spatial random functions). This means that the **covariance** $\text{COV}[Z(t_1), Z(t_2)] \neq 0$. Note that the further apart the points (e.g. t_1 and t_2) the lower the **covariance**. This is estimated as

$$\begin{aligned}\text{COV}[Z(t_1), Z(t_2)] &= E[(Z(t_1) - \mu_Z(t_1))(Z(t_2) - \mu_Z(t_2))] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - \mu_Z(t_1))(z_2 - \mu_Z(t_2)) f_Z(z_1, z_2; t_1, t_2) dz_1 dz_2\end{aligned}$$

if $t_1 = t_2$, $\text{COV}[Z(t_1), Z(t_2)] = \sigma_Z^2(t)$.

For n different locations t_i where $i = 1, 2, \dots, n$, the probability measure that a realization of $Z(t)$ has the values between $z_1 < z(t_1) \leq z_1 + dz_1$,

$z_2 < z(t_2) \leq z_2 + dz_2, \dots, z_n < z(t_n) \leq z_n + dz_n$. The **associated multivariate pdf** $f_{Z(t_1), Z(t_2), \dots, Z(t_n)}(z_1, z_2, \dots, z_n; t_1, t_2, \dots, t_n)$

$$\begin{aligned}f_{Z(t_1), Z(t_2), \dots, Z(t_n)}(z_1, z_2, \dots, z_n; t_1, t_2, \dots, t_n) = \\ \lim_{\partial z_1, \dots, \partial z_n \rightarrow 0} \frac{Pr(z_1 < z(t_1) \leq z_1 + \partial z_1, z_2 < z(t_2) \leq z_2 + \partial z_2, \dots, z_n < z(t_n) \leq z_n + \partial z_n)}{\partial z_1 \partial z_2 \dots \partial z_n}\end{aligned}$$

The **multivariate pdf** of a **random function** is sometime referred to as **multipoint pdf**, because it relates to random variables at different points or locations. Note that a **random function** is fully characterized if the **multivariate pdf** for any set of points is known.

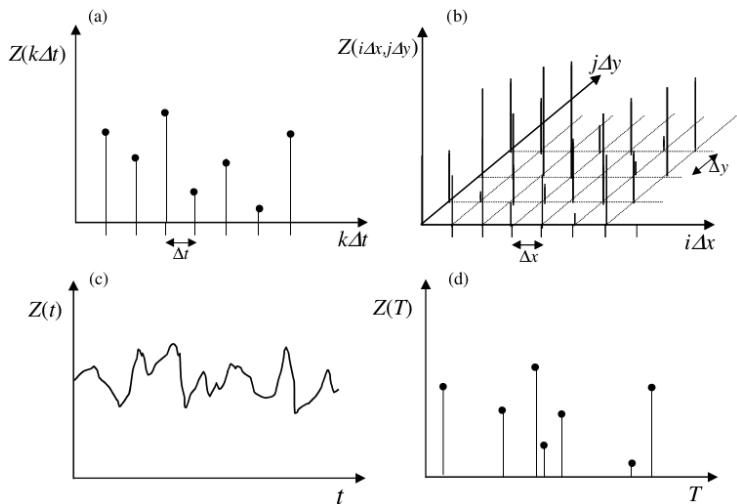
Types of random functions

Random functions can be classified as **continuous** whether the **random variable** is continuous (e.g. maximum annual river flows) or **discrete** whether the **random variable** is discrete (e.g. number of annual rainy days). Other way to classify a random function is based its domain (see figure). In the case of a **continuous random function**, this can be defined at:

1. **all locations** in time, space or space-time: $Z(t)$, $Z(\mathbf{x})$ or $Z(\mathbf{x}, t)$.
2. **discrete points** in time, space or space-time, where $Z(k\Delta t)$ ($k = 1, 2, \dots, K$) is referred to as a **random time series**, and $Z(i\Delta x, j\Delta y)$ ($i = 1, 2, \dots, I$; $j = 1, 2, \dots, J$) as a **lattice process**.
3. **random times** or **random locations** is space and time: $Z(T)$, $Z(\mathbf{X})$ or $Z(\mathbf{X}, T)$. This process is called **compound point process**. Note that the occurrence of the points at random coordinates in space and time is called a **point process**. If the occurrence of such a point is associated with the occurrence of a random variable (e.g. the occurrence of a thunder storm cell at a certain location in space with a random intensity of rainfall) it is called a compound point process.

The same distinction can be made for discrete-valued random functions: $D(t)$, $D(k\Delta t)$ or $D(T)$; $D(\mathbf{x})$, $D(i\Delta x, j\Delta y)$ or $D(\mathbf{X})$.

Types of random functions



This figure shows the different types of **random functions** based on the way they are defined on the functional domain: a) random time series, b) lattice process, c) continuous-time random function and d) compound point process.

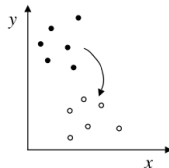
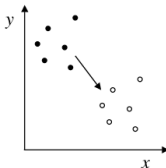
Stationary random functions

Strict stationary random functions

An special type of random functions are the **strict stationary random functions**. A **strict stationary random functions** is a random function for which its **multivariate pdf** is invariant under translation. This means that for any set of N locations and for any translations in time Δt , we have

$$f(z_1, z_2, \dots, z_N; t_1, t_2, \dots, t_N) = f(z_1, z_2, \dots, z_N; t_1 + \Delta t, t_2 + \Delta t, \dots, t_N + \Delta t) \quad \forall t_i, \Delta t$$

This assume that a configuration of points on the time axis can move backward and forward in time and the **multivariate pdf** does not change. For the spatial domain, **strict stationarity** means no spatial changes in the **pdf** if points are translated but no rotated. For instance a **strict random function** in 2D (see figure) the two sets of locations in the left-hand figure have the same **multivariate pdf**, while the locations in the right-hand figure necessarily does not. If the **multivariate pdf** of the **random function** is invariant under rotation, this is called a **statistically isotropic random function**.



Stationary random functions

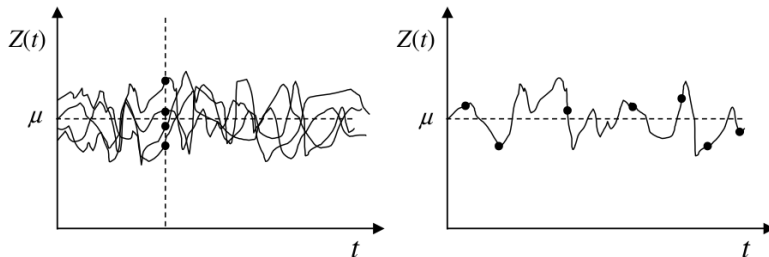
Ergodic random functions

- ▶ To estimate the statistical properties of a **random function**, a large number of realizations must be drawn.
- ▶ This is the case, for instance, of throwing a dice, from where one can generate multiple realizations.
- ▶ In reality this is not possible because we only have one realization: **reality**. This means that all relevant statistics of the **random function** are estimated from a single realization.
- ▶ In general, **ergodicity** deal with **stationarity** in **random functions**.
- ▶ To analyse **ergodicity**, statistical properties such as the **mean**, the **standard deviation** and the **covariance** are examined.
- ▶ **Stationarity** tells that all these properties are invariant no matter where you are (e.g. in time).
- ▶ For instance, for multiple realizations at point t_1 of $Z(t)$, the normal procedure to compute $\mu_Z(t_1)$ would be to average these realizations. However, this is impossible because in reality we only have one realization.
- ▶ If the **random function** is stationary the **pdf** $f_Z(z; t)$ at any location (in time) is the same and thus the statistical properties.
- ▶ This also means that within any single realisation we have at every location a sample from the same **pdf** $f_Z(z; t)$.

Stationary random functions

Ergodic random functions

- ▶ So, the **mean** can also be estimated if we take a sufficient number of samples from a single realisation, such as our **reality** (see figure below)



- ▶ This property of a **random function**, i.e. being able to estimate statistical properties of a random function from a large number of samples of a **single realisation** is called **ergodicity**.
- ▶ So for a **random function** is **ergodic** if it is **stationary**.
- ▶ Also for a **random function** to be **ergodic**, the samples from the single realisation should be taken from a large enough period of time or, in the spatial case, large enough area.

Stationary random functions

Ergodic random functions

A formal definition of **ergodicity** of a **random function** in its **mean**

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_T z(t) dt = \int_{-\infty}^{\infty} z f_Z(z; t) dz = \mu_Z$$

where T is an interval of time. This equation means that the integral over the probability distribution (the ensemble) for a given t (term on the left) can be replaced by a temporal (or spatial or spatio-temporal) integral of a very large interval T (or area or volume) (term on the right) to estimate μ_Z .

Similarly, **ergodicity** of a **random function** in its **covariance** is defined as

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T^2} \int_T z(t) z(t + \tau) dt &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - \mu_Z)(z_2 - \mu_Z) f(z_1, z_2; t, t + \tau) dz_1 dz_2 \\ &= \text{COV}[Z(T), Z(t + \tau)] \quad \forall \tau \end{aligned}$$

where τ is delta time.

Stationary random functions

Second order stationary random functions

A weaker form of **stationarity** is second order **stationarity**. This means that the bivariate (or two-point) **probability distribution** is invariant under translation

$$f(z_1, z_2; t_1, t_2) = f(z_1, z_2; t_1 + \tau, t_2 + \tau) \quad \forall t_1, t_2, \tau$$

In practice, one require only **ergodicity** and thus **stationarity** for the **mean**, **variance** and **covariance**. An **milder stationarity** is even considered and called **wide/weak sense stationarity**. For **wide sense stationary random functions**

$$\mu_Z(t) = \mu_Z \quad \text{constant and } t \text{ or } x \text{ independent}$$

$$\sigma_Z(t) = \sigma_Z \quad \text{constant and } t \text{ or } x \text{ independent}$$

$$\text{COV}[Z(t_1)Z(t_2)] = C_Z(t_2 - t_1) = C_Z(\tau) \quad \text{depends on } \tau \text{ (separation in time or space)}$$

The **covariance function** is a graph that describe the covariance and depends on τ (**lag**). The covariance function start when $t_1 = t_2$ (variance of both variables) and when τ increase the function tends to zero. This means that a random variable is not correlated for larger τ .

In the case of **random space functions** the **weak sense stationarity** means that $\text{COV}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = C_Z(\mathbf{x}_2 - \mathbf{x}_1) = C_Z(\mathbf{h})$, where $\mathbf{h} = \mathbf{x}_2 - \mathbf{x}_1$. In case of a **isotropic** random function $\text{COV}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = C_Z(|\mathbf{x}_2 - \mathbf{x}_1|) = C_Z(|\mathbf{h}|) = C_Z(h)$, where $|\mathbf{h}| = h$ is the **norm** of the difference vector.

Stationary random functions

Second order stationary random functions

When the **covariance function** is divided by the **variance** σ_Z^2 , this becomes the **correlation function** either $\rho_Z(\tau) = C_Z(\tau)/\sigma_Z^2$ or $\rho_Z(\mathbf{h}) = C_Z(\mathbf{h})/\sigma_Z^2$. When the data is regularly spaced either in time or space, the **covariance function** is:

$$\hat{C}_Z(k\Delta t) = \frac{1}{n-k} \sum_{k=1}^{n-k} (z_i - \hat{\mu}_Z)(z_{i+k} - \hat{\mu}_Z)$$

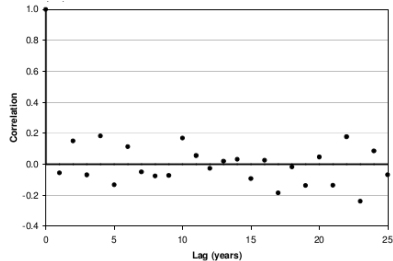
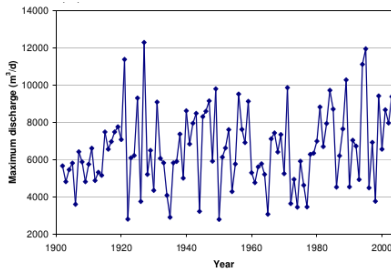
where $k\Delta t$ is the lag (in units of time or space) and Δt is the distance (in time or space) between observations. Frequently the estimator $\hat{C}_Z(k\Delta t)$ is used in the analysis of time series. In the case of irregularly positioned data in space, the **covariance function** is

$$\hat{C}_Z(\mathbf{h}) = \frac{1}{n(\mathbf{h})} \sum_{i=1}^{n(\mathbf{h})} (z(\mathbf{x}_i) - \hat{\mu}_Z)(z(\mathbf{x}_i + \mathbf{h} \pm \Delta \mathbf{h}) - \hat{\mu}_Z)$$

where $\mathbf{h}$ is the lag (a vector in space), $\Delta \mathbf{h}$ is a lag-tolerance which is needed to group a number of data-pairs together to get stable estimates for a given lag and $n(\mathbf{h})$ are the number of data pairs that are a distance $\mathbf{h} \pm \Delta \mathbf{h}$ apart.

Stationary random functions

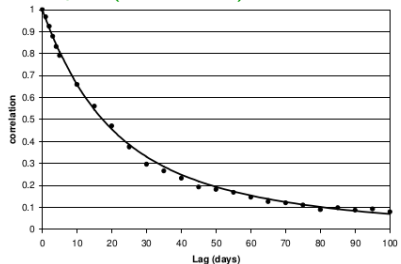
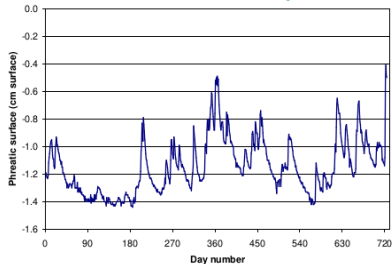
Correlation function: maximum annual streamflow



- ▶ The correlation plot, shows the **estimated** and the **fitted** correlation functions. According to these plots the data is uncorrelated.
- ▶ This uncorrelated behaviour of lagged data points is a characteristics of **white noise**, which are pure random processes, where the data points are uncorrelated no matter τ .
- ▶ **White noise** follow a **normal distribution**, so that, one can produce synthetic e.g. annual maximum streamflows after sampling the process. It would produce a time series with the same statistical properties of the original one.

Stationary random functions

Correlation function: daily water table depth (1994-1995)



- ▶ The correlation plot, shows the **estimated** and the **fitted** correlation functions. According to these plots the correlation decrease exponentially as τ increase.
- ▶ Again, one can produce synthetic e.g. daily water levels after sampling the process at the same discrete times. It would produce a time series with the same statistical properties of the original one.

Stationary random functions

Second order stationary random functions

The following table shows some models for modelling the **covariance function** for **isotropic random functions** and **wide sense stationary process**.

Exponential covariance	$C_Z(h) = \sigma_Z^2 e^{-h/a}$ for $h \geq 0, a > 0$
Gaussian covariance	$C_Z(h) = \sigma_Z^2 e^{-h^2/a^2}$ for $h \geq 0, a > 0$
Spherical covariance	$C_Z(h) = \begin{cases} \sigma_Z^2 \left[1 - \frac{3}{2} \left(\frac{h}{a} \right) + \frac{1}{2} \left(\frac{h}{a} \right)^3 \right], & \text{if } h < a \\ 0, & \text{if } h \geq a, h \geq 0, a > 0 \end{cases}$
Hole effect (wave) model	$C_Z(h) = \sigma_Z^2 \frac{b \sin(h/a)}{h}$ for $h \geq 0, a > 0$
White noise model	$\rho_Z(h) = \begin{cases} 1, & \text{if } h = 0 \\ 0, & \text{if } h > 0 \end{cases}$

Note that the **white noise model** has infinity variance, so it is not **wide sense stationary**.

Relations between various forms of stationarity

If a **random function** is **wide sense stationary** and its **multivariate pdf** is a **Gaussian distribution**, It is also second order stationary and also a strict sense **stationary random function**. More important, a wide sense stationary random function that is multivariate Gaussian (and thus also strict stationary) is completely characterised by only a few statistics: a constant **mean** $\mu_Z(t) = \mu_Z$ and a **covariance function** $C_Z(t_2 - t_1)$ that only depends on the separation distance.

Stationary random functions

Intrinsic random functions

An **Intrinsic random function** is a form of an **stationary random function**. For an intrinsic random function, we require (we show the spatial form here) that $E[Z(\mathbf{x}_2) - Z(\mathbf{x}_1)] = 0 \forall \mathbf{x}_2, \mathbf{x}_1$ and $E[Z(\mathbf{x}_2) - Z(\mathbf{x}_1)]^2 = 2\gamma_Z(\mathbf{x}_2 - \mathbf{x}_1) \forall \mathbf{x}_2, \mathbf{x}_1$. This means that the **mean** is constant and the **expected quadratic difference** is a function of the **lag vector** $\mathbf{h} = \mathbf{x}_2 - \mathbf{x}_1$, where the function $\gamma_Z(\mathbf{x}_2 - \mathbf{x}_1)$ is the **semivariogram**, which is defined as

$$\gamma_Z(\mathbf{x}_2, \mathbf{x}_1) = \gamma_Z(\mathbf{x}_2 - \mathbf{x}_1) = 1/2E[Z(\mathbf{x}_2) - Z(\mathbf{x}_1)]^2$$

The **semivariogram** can be estimated from observations as:

$$\hat{\gamma}_Z(\mathbf{h}) = \frac{1}{2n(\mathbf{h})} \sum_{i=1}^{n(\mathbf{h})} [z(\mathbf{x}_i) - z(\mathbf{x}_i + \mathbf{h} \pm \Delta\mathbf{h})]^2$$

Stationary random functions

Intrinsic random functions

Continuous **semivariograms** such the ones shown in the table can be fitted to the estimated **semivariogram** $\hat{\gamma}_Z(\mathbf{h})$.

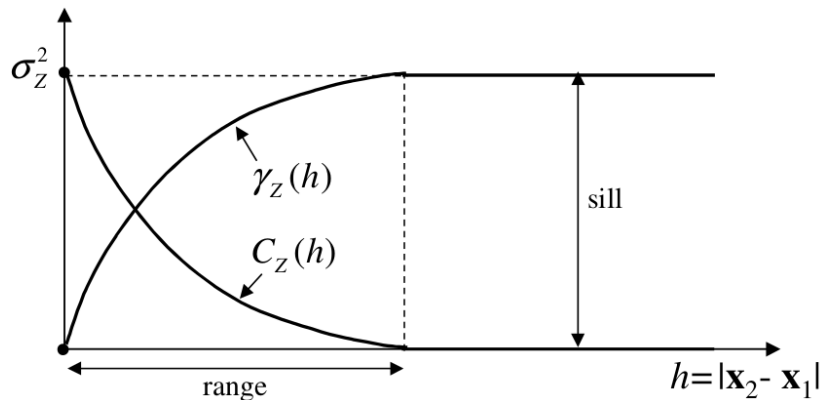
Exponential model	$\gamma_z(h) = c[1 - e^{-(h/a)}] \quad h \geq 0; a, c > 0$
Gaussian model	$\gamma_z(h) = c[1 - e^{-(h^2/a^2)}] \quad h \geq 0; a > 0$
Spherical model	$\gamma_z(h) = \begin{cases} c \left[\frac{3}{2} \left(\frac{h}{a} \right) + \frac{1}{2} \left(\frac{h}{a} \right)^3 \right] & \text{if } h < a \\ c & \text{if } h \geq a \end{cases}$ $h \geq 0; a, c > 0$
Hole effect (wave) model	$\gamma_z(h) = c \left[1 - \frac{b \sin(h/a)}{h} \right] \quad h \geq 0; a, c > 0$
Pure nugget model	$\gamma_z(h) = \begin{cases} 0 & h = 0 \\ c & h > 0 \end{cases} \quad c > 0$
Power model	$\gamma_z(h) = ah^b$ $h \geq 0; a > 0; 0 \leq b \leq 2$

where $c = \sigma_Z^2$ the variance of the stationary process.

Stationary random functions

Intrinsic random functions

Note that for an stationary random function, the **semivariogram** and the **covariance** functions are related as $\gamma_Z(\mathbf{x}_2 - \mathbf{x}_1) = \sigma_Z^2 - C_Z(\mathbf{x}_2 - \mathbf{x}_1)$ (see the figure).



Note that when the $C_Z \rightarrow 0$ for large $\mathbf{h} = \mathbf{x}_2 - \mathbf{x}_1$, γ_Z reach a plateau (the **zill**) which is equal to the **variance** σ_Z^2 . The distance at which this occurs is called the **range** of the **semivariogram**; beyond this distances, values on the random functions are no longer correlated.

Stationary random functions

Integral scale and scale of fluctuation

The **integral scale** or **correlation scale** is a measure of the degree of correlation for stationary random processes and it is defined as the area under the correlation function

$$I_{Z(t)} = \int_0^{\infty} \rho(\tau) d\tau$$

For the **exponential**, **Gaussian** and **spherical** correlation functions the integral scales are equal to a , $\frac{a\sqrt{\pi}}{2}$ and $\frac{3}{8}a$, respectively. Given that the correlation functions of **wide sense stationary processes** are even functions ($\rho(\tau) = \rho(-\tau)$), another measure of correlation is the **scale of fluctuation** defined as:

$$\theta = \int_{-\infty}^{\infty} \rho(\tau) d\tau = 2I_{Z(t)}$$

For random function in 2D and 3D, the **integral scales** are:

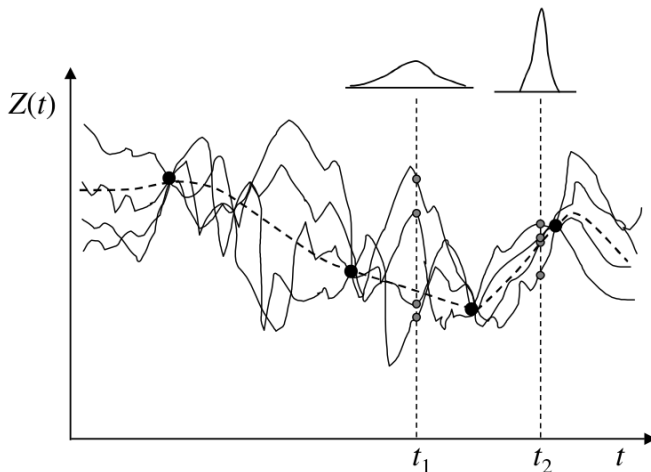
$$I_{Z(x)} = \left[\frac{4}{\pi} \int_0^{\infty} \int_0^{\infty} \rho(h_1, h_2) dh_1 dh_2 \right]^{1/2}$$

$$I_{Z(x)} = \left[\frac{6}{\pi} \int_0^{\infty} \int_0^{\infty} \int_0^{\infty} \rho(h_1, h_2, h_3) dh_1 dh_2 dh_3 \right]^{1/3}$$

Conditional random functions

Conditional random functions

Suppose that we have a **stationary random function** in time that is observed at a number of locations. Suppose for a moment that these observations are true (no errors at all). The figure shows multiple realizations of a **continuous time random function** that is observed at four location with no errors.



Conditional random functions

Conditional random functions

- ▶ Note that the realization are free to vary and differ between locations but are **conditioned** by these points, which is noticeable when comparing the **pdfs** at two locations t_1 and t_2 .
- ▶ It can be seen that uncertainty is larger further from an observation (t_1) than close to an observation (t_2).
- ▶ This is intuitively correct because an observation is able to reduce uncertainty for a limited interval proportional to the integral scale of the random function.
- ▶ At a distance larger than the range, the random values are no longer correlated with the random variable at the observation location and the uncertainty is as large (the variance of the pdf is a large) as that of the random function without observations.

Conditional random functions

Conditional random functions

- ▶ The random function that is observed at a number of locations and/or times is called a **conditional random function** and the probability distributions at locations t_1 and t_2 **conditional probability density functions (cpdfs)**: $f_Z(z; t_1|y_1, \dots, y_m)$ and $f_Z(z; t_2|y_1, \dots, y_m)$, where $y_1, \dots, y_m$ are the observations.
- ▶ The complete conditional random function is defined by the **conditional multivariate pdf**: $f(z_1, z_2, \dots, z_N; t_1, t_2, \dots, t_N|y_1, \dots, y_m)$.
- ▶ An alternative way of obtaining all the required statistics of the **conditional random function** is called **stochastic simulation**.
- ▶ The **conditional distribution** of $Z(s_1, t)$, or its **mean** value (see dashed line) and **variance**, can also be obtained directly through **geostatistical prediction** or **kriging** and **state-space prediction methods** such as the **Kalman filter**.